ENERGY LAB

CRASH TEST #05

PRESENTER GUIDE

English for Engineering — Unit 5: Energy (Career Paths: Engineering, Book 1, Lloyd & Frazier, Express Publishing 2011, pp. 12–13)

A 105-minute interactive lesson for ≈ 90 first-year engineering students, delivered by 8 student presenters (P1–P8) as one continuous engineering mission.

Total planned duration	105 minutes (see Section 2 for the minute-by-minute breakdown)
Slides	88 total · 88 visible · 0 hidden backup
Presenters	P1 Mission Commander · P2 Energy Engineer · P3 Dynamics Engineer · P4 Research Engineer · P5 Crash Investigator · P6 Communication Engineer · P7 Testing Engineer · P8 Chief Engineer
Companion files	ENERGY_LAB_UNIT5.pptx/.pdf · ENERGY_LAB_STUDENT_DOSSIER.pdf · quiz_questions.md · extras/team_selector.html · README.md

SECTION 1

Lesson overview

ENERGY LAB — CRASH TEST #05 turns Unit 5 (Energy) into one continuous engineering mission instead of eight separate mini-talks. The audience becomes a group of engineers, split into 8 companies (Alpha–Hotel). Across 105 minutes they: understand potential/kinetic energy → learn the engineering vocabulary (Tasks 3–4) → investigate the stretching-seatbelt test (the unit's reading abstract) → learn the language needed to discuss it (comparatives, be going to, present continuous for future, can, passive) → listen to two engineers (Tasks 6–7) → pitch a safer seatbelt design (Task 8) → write an engineering report (Task 9) → pass a final sequencing challenge.

The deck alternates TEACH → INTERACT → TEACH → CHALLENGE roughly every 8–15 minutes (four short games plus the Final Boss), scored with a simple Energy Points (EP) system: +100 EP for a correct team answer, +300 EP for a major challenge, and double EP on the Final Boss. Every slide's speaker notes carry PRESENTER / TIME / OBJECTIVE / SAY / ASK / EXPECTED ANSWER / ACTION / TRANSITION fields — open Presenter View in rehearsal so you always know what to say and where you're going next.

Source-fidelity rule used throughout this guide and the deck

Every fact tagged **TEXTBOOK** is quoted or minimally reworded from Career Paths: Engineering pp. 12–13. Everything tagged **SUPPLEMENTARY** (games, extra revision statements, illustrative sentences, the 6-question reading scaffold) is teaching support written for this lesson — useful, but never presented to students as something the textbook itself says. Section 6 and Section 13 mark each item.

SECTION 2

Exact 105-minute schedule

Time	Duration	Segment	Presenter(s)
00:00–00:05	5 min	**COLD OPEN**	P1
00:05–00:12	7 min	**POTENTIAL VS KINETIC**	P2
00:12–00:18	6 min	**GAME 1 · ENERGY SCAN**	P2
00:18–00:30	12 min	**ENGINEERING VOCABULARY**	P3, P4
00:30–00:39	9 min	**GAME 2 · VOCABULARY BATTLE**	P4
00:39–00:42	3 min	**CRASH ALERT**	P1
00:42–00:56	14 min	**SEATBELT INVESTIGATION**	P5
00:56–01:03	7 min	**GAME 3 · CRASH DETECTIVE**	P5
01:03–01:16	13 min	**LANGUAGE OF ENGINEERS**	P6
01:16–01:22	6 min	**GAME 4 · ERROR 404**	P6
01:22–01:30	8 min	**LISTENING**	P7
01:30–01:38	8 min	**DESIGN CHALLENGE / SPEAKING**	P7
01:38–01:42	4 min	**WRITING**	P8

Time	Duration	Segment	Presenter(s)
01:42–01:45	3 min	**FINAL BOSS + CONCLUSION**	P1

Total planned time: **105 minutes** (fits the 105-minute slot exactly; see Section 19 for rehearsal timing checks). Act boundaries follow the deck's own progress bar (bottom of every slide).

SECTION 3

Presenter assignments (P1–P8) and slide ranges

Slide numbers below match the PPTX exactly (open View → Slide Navigator to jump).

Presenter	Role	Focus	Slides
P1	MISSION COMMANDER	Opening, narrative, transitions, final conclusion, time management	1–5
P2	ENERGY ENGINEER	Potential energy, kinetic energy, Energy Scan activity	6–20
P3	DYNAMICS ENGINEER	Decelerate, stopping distance, force, work	21–26
P4	RESEARCH ENGINEER	Transfer, G, work-energy principle, abstract, vocabulary battle	27–32
P1	MISSION COMMANDER	Opening, narrative, transitions, final conclusion, time management	33–34
P5	CRASH INVESTIGATOR	Main reading, seatbelt experiment, comprehension, True/False challenge	35–55
P6	COMMUNICATION ENGINEER	Useful language / grammar, language correction challenge	56–72
P7	TESTING ENGINEER	Listening, speaking, engineering pitch	73–81
P8	CHIEF ENGINEER	Writing, final assessment support	82–83
P1	MISSION COMMANDER	Opening, narrative, transitions, final conclusion, time management	84–88

P1 (Mission Commander) reappears at the Crash Alert transition and the Final Boss/closing — this is intentional: P1 is the narrator who frames every act change, not just the opener.

SECTION 5

Learning objectives

By the end of the lesson, students should be able to:

- Define and distinguish **potential energy** and **kinetic energy**, and explain how one becomes the other in a crash.
- Use the ten Unit 5 vocabulary items accurately: decelerate, stopping distance, kinetic/potential energy, force, work, abstract, work-energy principle, G, transfer.
- Explain the three-part structure of a scientific abstract (Background / Results / Conclusions) and summarise the Stretching Seatbelt Test using it.
- Answer literal and inferential comprehension questions on the reading (textbook Task 2 True/False).

- Use comparatives (safer than…), be going to, present continuous for future arrangements, can + base verb, and the passive (be + past participle) to talk about the engineering design.
- Understand a short listening exchange between two engineers and complete a gapped transcript (Tasks 6–7).
- Deliver a short spoken pitch (Task 8) and write a 100–120-word engineering report (Task 9) covering design, energy transfer, and safety.

SECTION 6

Complete vocabulary definitions

Task 3 (six items) and Task 4 (four items) — all definitions/examples below are textbook-sourced except the plain-English gloss column, which is teaching support.

Term	Task	Textbook def. / example	Plain gloss (support)
decelerate	Task 3.1	"Drivers decelerate when they press the brakes."	to reduce speed (to slow down)
stopping distance	Task 3.1	"The stopping distance of this car is dangerously long."	the distance an object travels before it stops
kinetic energy	Task 3.2	"A moving object has kinetic energy."	the energy of motion
potential energy	Task 3.2	"A still object has potential energy."	stored energy (an object at rest, in position)
force	Task 3.3	"When you kick a ball, the force moves it forward."	a push or a pull that makes an object move
work	Task 3.3	"A machine does work when it moves another object."	what a force does when it moves an object
abstract	Task 4.1 = A	"a summary of an article, document, or other text"	a short summary at the start of a research text
work-energy principle	Task 4.2 = D	"the idea that a moving object's energy equals the distance it moves"	links energy, force and distance
G	Task 4.3 = C	"the unit of force that equals the force of gravity"	a unit used to measure how strong a force feels
transfer	Task 4.4 = B	"to move from one object to another"	here: energy moves from the passenger to the belt

SECTION 7

Reading explanation — the Stretching Seatbelt Test abstract

TEXTBOOK — Abstract, quoted in full (Background / Results / Conclusions):

> *Passengers who are riding in a car have potential energy. In a crash, a car decelerates quickly. This causes the passengers' potential energy to become kinetic energy. Passengers wearing seatbelts stop. With non-stretching seatbelts, the passenger's stopping distance is short. Stretching seatbelts increase this distance. The work-energy principle shows that a slightly greater stopping distance exerts less force on passengers.*
>
> *Tests show that stretching seatbelts reduce the force and Gs that passengers feel.*

The new seatbelts reduce injuries. This is because less work is done on passengers. Instead, most of the passenger's kinetic energy transfers to the belt.

Why the abstract is structured this way (teaching support): Background = why the test was run; Results = what was measured; Conclusions = what it means. This is a universal research-writing convention, not specific to this textbook — teach it once and students can read any abstract afterwards.

SUPPLEMENTARY — six-question reading scaffold (given to students before they read; maps onto the abstract above, used to prepare for textbook Task 2):

#	Question	Answer (from the abstract above)
1	What happens when the car decelerates?	In a crash, a car decelerates quickly.
2	What happens to the passenger?	The passenger's potential energy becomes kinetic energy — the passenger keeps moving until stopped.
3	What does the new seatbelt do?	It stretches (stretching seatbelts).
4	What happens to stopping distance?	It increases: a slightly greater stopping distance.
5	What happens to force / Gs?	They decrease: less force, fewer Gs.
6	What happens to injuries?	They are reduced.

TEXTBOOK Task 2 — True/False (items 1–3) + SUPPLEMENTARY revision statements (items 4–7), used in Game 3 — see Section 13 for the full answer key.

SECTION 8

Grammar / language explanation

Structure	Pattern	Textbook example	Notes
Comparatives	long → longer · short → shorter · safe → safer · great → greater	“The new seatbelt is safer than the current seatbelt.”	Short adjectives take -er (not ‘more + adjective’). Pattern: adjective + ER + THAN.
Be going to	SUBJECT + BE + GOING TO + BASE VERB	“Are you going to make the seatbelts stretch more?” (Task 8 language box)	Meaning: a plan / intention, already decided, no fixed time.
Present continuous (future)	BE + VERB-ING (+ a future time expression)	“We're testing it next week.” (Task 7/8)	Meaning: a fixed arrangement — contrast with be going to (plan, no fixed time) using the same two textbook sentences.
Can	CAN + BASE VERB (never ‘can to’ / never ‘can + -s’)	“How can you do that?” (Task 8 language box)	Correct: can make, can reduce, can transfer. Incorrect: can to make, can reduces.
Passive	BE + PAST PARTICIPLE · (future: WILL + BE + PAST PARTICIPLE)	“Work is done on passengers.” (Abstract) · “Energy will be transferred to the belt.” (Task 7)	Used because the focus is on the passenger/energy, not on who acts.

Game 4 (Error 404) drills all five structures as error-correction — answer key in Section 13.

SECTION 9

Listening instructions (Tasks 6–7)

Before playing anything: confirm whether you have the official Express Publishing Unit 5 audio. This deck never fabricates a transcript beyond what is printed in the textbook. If the official audio is unavailable, use listening_fallback.md (two presenters read the printed dialogue live, slowly, at natural stress/intonation) — see Section 16.

Task 6 — two multiple-choice questions (play once, no writing; reveal both together):

1. What is the conversation mostly about?

- A) the drawbacks of longer stopping distances
- B) the benefits of a new seatbelt design ← ANSWER
- C) the safety of non-stretching seatbelts
- D) the failure of a new seatbelt design

2. The woman thinks the new seatbelt will…

- A) fail the test next week.
- B) transfer potential energy.
- C) shorten the stopping distance.
- D) be safer than the current seatbelts. ← ANSWER

Task 7 — complete the conversation (6 gaps; play again, allow writing):

Speaker	Line
Engineer 1	I heard you're working on the new seatbelt designs.
Engineer 2	Yes, I am. I think I can make them [1] our current models.
Engineer 1	Really? How can you do that?
Engineer 2	All I have to do is extend the [2].
Engineer 1	How? Are you going to make the seatbelts stretch more?
Engineer 2	Yes, exactly. That way, the passenger's [3] will be transferred to the [4].
Engineer 1	That's a [5] idea.
Engineer 2	Yeah, I think it is. We're [6] it next week.

Gap	Answer
1	safer than
2	stopping distance
3	kinetic energy
4	belt
5	(from official audio — not printed in the textbook)
6	testing

Gap 5 note: two-word positive phrase — NOT printed in the book; take it from the official audio. Confirm it from the official audio/answer key before class; do not present a guess to students as the textbook's confirmed answer.

SECTION 10

Speaking activity instructions (Task 8 — Board Meeting)

Textbook roles (Task 8): **Student A** has a new seatbelt design and talks to Student B about safety, stopping distance, and energy transfer. **Student B** is an engineer who discusses the effects of the new design with Student A. In the lesson this becomes a whole-company “board meeting” pitch.

Each 45-second company pitch must include (supplementary checklist built from the unit's vocabulary + Task 8 roles):

- stopping distance
- kinetic energy
- transfer
- force
- one comparative
- one future structure (going-to or present continuous)

Closing question every company must answer: **“Why is your design safer?”** Give 4 minutes of prep (static timer slide + a visible phone timer — never rely on an automatic slide timer), then use extras/team_selector.html (or 8 folded paper slips) to pick 2–3 companies to present live in the 8 minutes available; the rest hand in written notes for a quick mentor check.

SECTION 11

Writing activity instructions (Task 9 — Engineering Report)

Textbook Task 9 (100–120 words), three focus points: how to extend a passenger's stopping distance; what effect this has on energy transfer; how this affects safety. The deck presents this as three paragraphs — Design / Energy / Safety — with connectors (First, When…, As a result, Therefore, This means that…). **Do not project a full model answer before students write** — it must be their own writing.

MODEL ANSWER (presenter preparation only — 109 words, faithful to the unit; keep this out of students' sight until after they submit):

> *Our new seatbelt design is safer than the current model. First, we can extend the passenger's stopping distance by making the seatbelt stretch more. When a crash happens, the car decelerates quickly and the passenger keeps moving. With a stretching seatbelt, most of the passenger's kinetic energy will be transferred to the belt. As a result, less work is done on the passenger, and a slightly greater stopping distance exerts less force. Therefore, passengers feel fewer Gs. This means that the new seatbelt reduces injuries and protects passengers better than a non-stretching seatbelt. We are testing the new design next week, and we are going to report the results.*

SECTION 12

Game rules

Game 1 — Energy Scan

Presenter shows a situation. Students respond with FIST = potential energy, OPEN HAND = kinetic energy, on “3·2·1·SHOW!”. Mentors award +100 EP per zone that is mostly correct. 5 situations; the 5th (ball rolling down a ramp) is deliberately harder — accept fist-then-open-hand or “both”.

Game 2 — Online Vocabulary Battle

8-question live quiz (Wayground/Quizizz/Kahoot-style) on the 10 vocabulary words — see quiz_questions.md. QR code and PIN on the deck are placeholders; replace before class. No-Wi-Fi fallback: read the same 8 questions aloud, teams answer with A/B/C/D fingers or cards, mentors award +100 EP.

Game 3 — Crash Detective

True/False, THUMBS UP = true / THUMBS DOWN = false, on 7 statements (3 from textbook Task 2, 4 supplementary revision statements built on the abstract). +100 EP per zone that is mostly correct.

Game 4 — Error 404: Engineer English

5 broken sentences built on the 5 grammar points just taught; students say the correct version aloud. Fastest correct team per zone: +100 EP.

Final Boss

Whole-class sequencing challenge (4 cards) worth DOUBLE EP — see Section 13 for the correct order.

Scoring stays deliberately simple: +100 EP correct team answer, +300 EP major challenge (if you choose to mark one game as “major”, e.g. Game 2), double EP on the Final Boss. Do not add sub-rules — mentors should be able to award EP in under 5 seconds.

SECTION 13

Correct answers (consolidated key)

Task 3 (fill the blanks)

decelerate / stopping distance

- The _____ of this car is dangerously long. → **stopping distance**
- Drivers _____ when they press the brakes. → **decelerate**

kinetic energy / potential energy

- A still object has _____. → **potential energy**
- A moving object has _____. → **kinetic energy**

force / work

- When you kick a ball, the _____ moves it forward. → **force**
- A machine does _____ when it moves another object. → **work**

Task 4 (match)

- 1 = abstract — A
- 2 = work-energy principle — D
- 3 = G — C
- 4 = transfer — B

Game 3 — True/False (7 statements)

#	Statement	Answer	Why / source
1	Kinetic energy becomes potential energy in a crash.	FALSE	The abstract says the opposite: the passengers' POTENTIAL energy becomes KINETIC energy. (Task 2.1 · textbook)
2	Shorter stopping distances exert less force on people.	FALSE	A slightly GREATER stopping distance exerts LESS force. Shorter distance = more force. (Task 2.2 · textbook)
3	Stretching seatbelts cause less work to be done on people.	TRUE	Conclusions: 'less work is done on passengers'. (Task 2.3 · textbook)
4	Stretching the seatbelt increases the stopping distance.	TRUE	Background: 'Stretching seatbelts increase this distance.' (Revision · supplementary)
5	The tests showed an increase in Gs.	FALSE	Results: stretching seatbelts REDUCE the force and Gs that passengers feel. (Revision · supplementary)
6	With non-stretching seatbelts, the stopping distance is long.	FALSE	Background: with non-stretching seatbelts the stopping distance is SHORT. (Revision · supplementary)
7	Most of the passenger's kinetic energy transfers to the belt.	TRUE	Conclusions: 'most of the passenger's kinetic energy transfers to the belt'. (Revision · supplementary)

Listening — Task 6/7: Q1 = B, Q2 = D. Gap-fill answers: 1=safer than, 2=stopping distance, 3=kinetic energy, 4=belt, 6=testing. (Gap 5 — confirm from official audio.)

Game 4 — Error 404 fixes

- **We can make it safer.**
- **The new seatbelt is safer.**
- **We're going to test it.**
- **We're testing it next week.**

- **Energy will be transferred to the belt.**

Final Boss — correct order

B (SEATBELT STRETCHES) → D (STOPPING DISTANCE INCREASES) → A (FORCE ON PASSENGER DECREASES) → C (INJURIES DECREASE)

SECTION 14

Transition scripts

Every slide's speaker notes include a TRANSITION line bridging to the next slide/presenter — use those in rehearsal for the moment-to-moment handoffs. The two book-ended scripts below carry the whole lesson's frame story.

OPENING SCRIPT (P1, cold open with a real ball)

> *Good morning everyone. Before we start, I need one volunteer. [Volunteer comes forward.] Is this ball moving? [No.] Does it have energy? [Allow answers.] [Raise the ball.] What about now? [Allow answers.] [Drop/roll it safely.] And now? [Allow answers.] Interesting. Keep your answers in mind. By the end of today's session, you'll be able to explain exactly what is happening. [Reveal ENERGY LAB.] For the next hundred minutes, you're not simply students. You're engineers.*

CLOSING SCRIPT (P1, same ball returns)

> *One hour and forty-five minutes ago, we showed you this ball and asked one simple question: does it have energy? Now you can distinguish potential and kinetic energy. You can explain what happens in the seatbelt experiment. You can use the engineering vocabulary. You can discuss an engineering design in English. And most importantly, you didn't just study Unit 5. You engineered it. Thank you, engineers.*

SECTION 15

Materials checklist

- A real ball (used in the cold open and the closing) — something safe to drop/roll indoors.
- Presenter laptop with ENERGY_LAB_UNIT5.pptx, fully charged + charger.
- HDMI/USB-C adapter matching the classroom projector.
- PDF backup of the deck open in a second window/tab.
- extras/team_selector.html open in a browser tab, tested beforehand.
- Printed ENERGY_LAB_STUDENT_DOSSIER.pdf, one per student (or per pair).
- 8 zone signs (ALPHA–HOTEL) placed around the room before students arrive.
- Optional: A/B/C/D cards per student or per team, as a backup for any multiple-choice moment.
- Phone with a visible stopwatch/timer app (for the 3-minute reading and 4-minute pitch-prep windows).
- Official Unit 5 audio file/CD track, tested on the classroom speakers, if available.

SECTION 16

Backup plan — if classroom Wi-Fi / internet fails

The lesson must never stop because of connectivity. Two moments depend on the internet:

- **Game 2 (Online Vocabulary Battle):** skip the QR/PIN slide. Read the 8 questions from quiz_questions.md aloud; teams answer with A/B/C/D fingers or cards; mentors award +100 EP per correct zone.
- **Random company selection:** if extras/team_selector.html cannot load (it needs no internet once opened, but confirm the laptop/browser works), fold eight paper slips with the company names and draw one.

SECTION 17

Backup plan — if the official listening audio fails

Never invent or play a fake “official” recording. If the official Unit 5 audio is missing or the file/speakers fail: two presenters perform the printed Task 7 dialogue live, slowly, with clear stress on the six gapped words — full delivery notes are in listening_fallback.md. State to the class that this is the presenters reading the textbook dialogue, not a recording.

SECTION 18

Setup instructions before class

- Arrive early enough to test the projector resolution against the deck's 16:9 slides.
- Open ENERGY_LAB_UNIT5.pptx in PowerPoint (or Keynote) and check the fonts substituted correctly (Arial / Arial Black / Arial Narrow are all Mac-system fonts — no download needed).
- Run the deck once in Presenter/Rehearse mode to confirm every presenter can see their own speaker notes.
- Place the 8 zone signs (ALPHA–HOTEL) around the room.
- Open extras/team_selector.html in a browser tab and test the SELECT COMPANY button once.
- Load the official Unit 5 audio and test playback volume with the classroom speakers.
- Print and stack the student dossiers where they can be handed out quickly.
- Replace the Game 2 QR code / PIN with your real quiz platform link (see README.md).

SECTION 19

Rehearsal checklist

REHEARSAL #1 — Content accuracy

Each presenter studies their own slide range (Section 3) and speaker notes alone; confirm every presenter can explain their concept without reading the slide verbatim.

REHEARSAL #2 — Full run with a stopwatch

Run the entire 88-slide deck start to finish, out loud, with real handoffs between presenters. Target: ≤ 105 minutes. Note which sections ran long and shorten repeated examples or game question counts first — never cut the main reading, vocabulary, required listening, speaking task, writing explanation, or final recap.

REHEARSAL #3 — Technology rehearsal

- PPTX opens and displays correctly on the real classroom projector
- Audio/speakers
- QR code / live quiz platform (once you've replaced the placeholder)
- extras/team_selector.html
- PDF backup opens as a fallback

FINAL REHEARSAL — Perform exactly like class

No stopping, no “let's redo that bit.” Record the actual duration and compare against 105 minutes.

SECTION 20

Day-of-class checklist

- □ Ball, laptop + charger, adapter in bag
- □ Printed student dossiers counted out
- □ Zone signs placed before students arrive
- □ Waiting screen (slide 1) on display as students enter
- □ Projector + audio tested in the actual room, not just at home
- □ Game 2 QR/PIN replaced with the real quiz
- □ team_selector.html open and tested
- □ Official audio cued to the right track
- □ All 8 presenters present and know their slide range
- □ Phone timer ready for the 3-minute reading and 4-minute pitch prep

End of guide. See README.md for the day-of quick reference and ENERGY_LAB_STUDENT_DOSSIER.pdf for the printable student handout.